Upper high school · probability and sequences

Waiting for the first success

Lesson and method

Before you begin

Independent trials, geometric sequences and sums.

Ideas and method

Repeat independent trials, each with success probability p, where $0 < p \leq 1$. T is the trial number of the first success, starting at 1.

The event $T = k$ requires $k - 1$ failures followed by success: $P(T = k) = (1 - p)^{k-1}p$. Likewise, $P(T > n) = (1 - p)^n$ for $n \geq 0$.

The expectation is $E(T) = \frac{1}{p}$. It follows from the tail-sum formula $E(T) = \sum P(T > n)$, for $n \geq 0$, and the geometric series. This is an average, not a guaranteed success time.

Worked example

For a fair die with success defined as rolling $6, p = \frac{1}{6}$. The mean waiting time is 6 rolls, but the probability of waiting more than 6 rolls is $(\frac{5}{6})^6$, about 0.335.

Explore

For $p = \frac{1}{4}$, compare $P(T > n)$ for $n = 0, 1,...,10$. Relate the decrease to the geometric factor $\frac{3}{4}$.